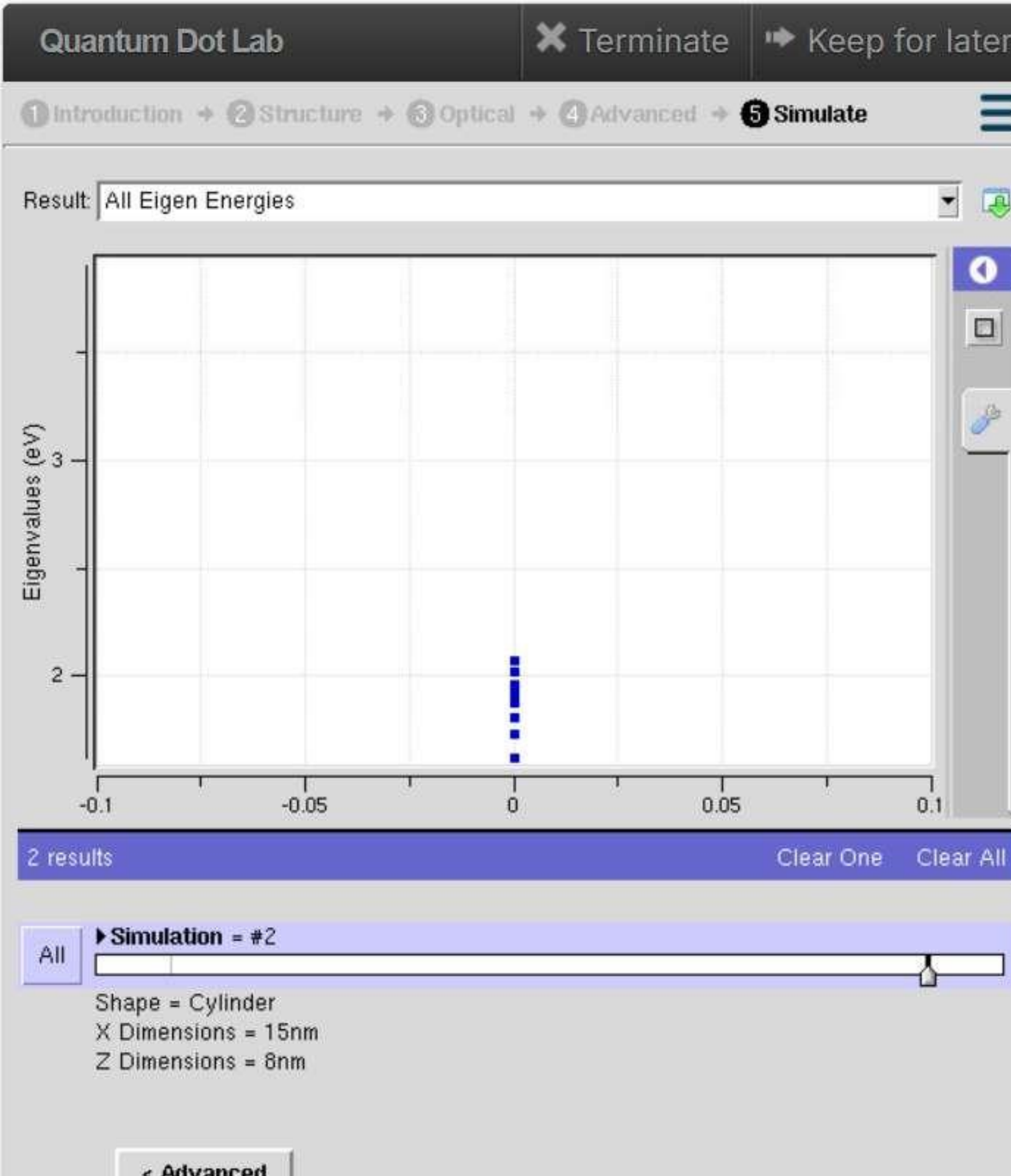
ECA3102
Nanoelectornics
Exp2: Simulation of Energy States of Quantum Dot using NanoHub
Aim: To simulate the **energy states of a semiconductor quantum dot** using the **NanoHUB** simulation platform and analyze the effect of three-dimensional quantum confinement on the discrete energy levels.
Case Study 1: Simulate the energy states of cylindrical quantum dots of X=15 nm, Y = 10.5nm and Z = 8 nm.
Aim: To simulate the **energy states of a cylindrical quantum dot** with dimensions **X = 15 nm, Y = 10.5 nm, and Z = 8 nm** using NanoHUB. The study aims to analyze the effect of three-dimensional quantum confinement on the discrete electron energy levels and understand the electronic behavior of nanoscale semiconductor structures.



Case Study 2: Simulate the energy states of two layer cuboid quantum dots

Aim: To simulate the **energy states of a two-layer cuboid quantum dot** using NanoHUB and analyze the effect of multilayer quantum confinement on the discrete electron energy levels.

Why: The study aims to investigate how the layered structure influences the electronic properties for potential nanoelectronic and optoelectronic device applications.